

# TIFFANY TAN

## Applied AI Engineer

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### SUMMARY

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Seven years across supply chain, aviation, and healthcare AI. What keeps me here is building AI with tangible impact — production systems people actually rely on — and the relationships I get to build with people along the way. Loves hiking and mother nature.

### SKILLS

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- Computer Vision: classification & detection (ResNet, YOLO, MobileNet), segmentation (SAM2), self-supervised pretraining (autoencoders), medical image preprocessing & alignment, keypoint/pose estimation (MediaPipe), 3D reconstruction (Structure-from-Motion, COLMAP), model evaluation (mAP, F1, ROC-AUC, ablation studies)
- AI/ML: PyTorch, LLMs & agentic AI (LangChain, Copilot Studio, OpenAI/Anthropic APIs), NLP (Transformers)
- Product: roadmap & sprint planning (Azure DevOps), user adoption analytics, stakeholder & clinician engagement, build-vs-buy evaluation, responsible AI, budget ownership
- Data & Cloud: Python, SQL, Spark, Databricks (Delta Lake), Microsoft Azure, FastAPI, Docker, CI/CD, Power BI

### WORK EXPERIENCE

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#### Louis Dreyfus Company — *Product Manager / Applied AI Engineer, AI Hub*

*Jun 2025 – Present*

- Product Manager for an internal vessel tracking platform serving 1,000+ monthly active users across 20 countries, driving cross-platform adoption and AI feature integration; own US\$500K annual product budget
- Lead cross-functional team of 5 across sprint planning, user feedback-driven roadmaps, and CI/CD (Azure DevOps); ship Python services via FastAPI and architect Delta Lake data integrations for near-real-time intelligence
- Shipped a conversational AI vessel tracking agent in Microsoft Teams (Copilot Studio) with multi-vessel query support, extending the platform into agentic AI workflows
- Built a YOLO-based object detection system for palm fresh fruit bunch (FFB) detection and ripeness classification from plantation imagery (83% F1), trained and deployed on Azure ML GPU infrastructure to automate fruit counting and yield forecasting for trading and harvest planning
- Automated subjective manual crop surveys into a repeatable, auditable AI-driven yield forecasting process with projected US\$1M+ annual commercial margin improvement, saving ~140 minutes per surveyor daily (~64 man-days/year); established dataset curation and labeling workflows and secured charter approval across AI hub and Indonesia plantation teams
- Drive build-vs-buy evaluations for external data partnerships (port intelligence, vessel lineup APIs), balancing vendor cost against internal automation

#### SATS Ltd — *Data Scientist, Assistant Manager*

*Jan 2024 – Jun 2025*

- Lead data scientist for end-to-end object detection pipeline (YOLO, MobileNet) for missing shipment tracking: 50% reduction in missing shipments, S\$150K annual savings; scaled PoC to production in 10 months with Docker, Azure, and CI/CD, optimizing inference via quantization and batching
- Architected multimodal AI system combining an LLM-powered chatbot (LangChain, OpenAI API) with image classification for automated incident severity detection, orchestrated with Airflow — cutting incident resolution time 20% and saving 100+ man-hours weekly

#### Johnson & Johnson — *APAC Data Analyst / Scientist, Healthcare Supply Chain*

*Dec 2021 – Dec 2023*

- Deployed demand forecasting models (ARIMA, Random Forest) for medical product portfolios, improving forecast accuracy 5% and reducing inventory 10%
- Automated reporting pipelines (Python, Alteryx, Power Query) and built Power BI dashboards, lifting reporting accuracy to 95% and improving portfolio visibility

#### Ninja Van — *Data Analyst, Regional Operations*

*Jul 2019 – Nov 2021*

- Partnered with regional COOs on data-driven operational improvements; optimized SQL pipelines (25% faster queries) and automated weekly reporting in Python

### RESEARCH

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#### KK Women's & Children's Hospital × NTU — *AI Researcher, Pediatric Fracture Classification*

*2026 – Present*

- Research collaborator on AI-assisted Gartland classification of pediatric supracondylar elbow fractures from X-rays, working directly with orthopedic clinicians and academic supervisors

- Designed a clinically aligned cascade of binary ResNet18 classifiers mirroring the surgical decision flowchart (AP and lateral views); benchmarked supervised vs. self-supervised (autoencoder) pretraining with systematic ablations across label smoothing, MixUp, and dropout
- Built automated X-ray alignment preprocessing (PCA orientation, Otsu thresholding, SAM2 bone segmentation) to standardize unaligned clinical images, and presented methodology and results to clinicians in interpretable terms

EDUCATION

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| <b>Nanyang Technological University — <i>MSc in Artificial Intelligence (GPA 4.67/5.00)</i></b>                                                                                                                                                    | <i>Aug 2024 – Dec 2026</i> |
| <ul style="list-style-type: none"><li>• IMDA postgraduate scholarship; coursework in computer vision, deep learning for NLP, LLMs, time series forecasting, AI ethics. Built a Structure-from-Motion 3D reconstruction pipeline (COLMAP)</li></ul> |                            |
| <b>General Assembly — <i>Data Science Immersive</i></b>                                                                                                                                                                                            | <i>Jun 2023 – Dec 2023</i> |
| <ul style="list-style-type: none"><li>• Capstone: real-time posture analysis for injury prevention using MediaPipe (97% accuracy)</li></ul>                                                                                                        |                            |
| <b>National University of Singapore — <i>BEng Chemical Engineering, Honours with Distinction</i></b>                                                                                                                                               | <i>Aug 2015 – Jun 2019</i> |
| <ul style="list-style-type: none"><li>• GPA 4.08/5.00; Captain, NUS Cross Country</li></ul>                                                                                                                                                        |                            |

CERTIFICATIONS

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Generative AI with LLMs, DeepLearning.AI (2024) | AI4I® Literacy in AI, AI Singapore (2024) | Power BI Data Analyst, Microsoft (2022)